Modeling Liquidity Spillover Dynamics Across Equity, Cryptocurrency, and Precious Metals Markets Using Econometric Methods

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***Abstract*—The interconnectedness of global financial markets means a liquidity shock in one asset class can cascade into others, yet no study has simultaneously measured this across equity, cryptocurrency, and precious metals markets. Existing work focuses on return and volatility spillovers, leaving liquidity transmission poorly understood. This paper applies the Amihud illiquidity ratio to daily data for six assets: S&P 500, NASDAQ, Bitcoin, Ethereum, gold, and silver, from January 2020 to December 2024. Vector autoregression with forecast error variance decomposition quantifies directional spillovers across all six markets simultaneously. Rolling 60-day correlations identify regime shifts across five market periods, and a liquidity-adjusted mean-variance portfolio penalizes illiquid assets in the optimization objective. The dominant own-market variance observed in the FEVD matrix confirms that each market primarily drives its own liquidity conditions, while meaningful off-diagonal transmission occurs especially between equity and precious metals markets. On the worst five percent of equity days, Bitcoin averages -3.3% while gold averages near zero, confirming gold as a weak safe haven and rejecting Bitcoin as one. The liquidity-adjusted portfolio achieves a Sharpe ratio of 1.15, equal to standard mean-variance, with the liquidity penalty steering allocation away from high-illiquidity assets. Future work will apply quantile-frequency decomposition to separate tail from median liquidity transmission.**

***Index Terms*—liquidity spillover, cross-market connectedness, Amihud ratio, VAR-FEVD, cryptocurrency, precious metals, safe-haven, portfolio optimization, financial contagion**

I. Introduction

Global financial markets changed in ways that most textbooks had not fully anticipated during 2020 and the years that followed. The COVID-19 pandemic triggered synchronized price declines across asset classes that had, until then, been considered largely independent. Gold, widely held as a crisis hedge, briefly sold off alongside equities in March 2020 as institutions liquidated assets to meet margin calls [10]. Bitcoin, which some analysts had called "digital gold," fell by over 50% in a matter of days [8]. Silver moved in step with equity indices rather than with gold. The usual diversification benefits simply disappeared.

What this episode exposed was not just return co-movement, but something deeper: liquidity dried up simultaneously across multiple markets. When investors needed to sell, buyers stepped back everywhere at once. This is the phenomenon this paper investigates: cross-market liquidity transmission, or how a shock to trading conditions in one market propagates into other markets.

Liquidity is not the same as volatility. A market can be highly volatile while remaining liquid: options expiry days, for instance. And a market can be calm on price terms while becoming profoundly illiquid, as gold was during parts of the 2020 crash. Volatility spillover models, which dominate the existing literature, miss this distinction entirely. The Amihud [1] illiquidity ratio, defined as the ratio of absolute daily return to dollar trading volume, captures this dimension directly. It measures price impact per unit of trading activity, a concrete proxy for how difficult it is to trade without moving the price.

Current approaches to cross-market connectedness fall into three broad categories. The Diebold-Yilmaz framework, built on forecast error variance decomposition from a vector autoregressive model, is the most widely adopted method for measuring directional spillovers between asset markets [2], [3]. Time-varying extensions using rolling windows or TVP-VAR have become standard for tracking how connectedness shifts across market regimes [11]. A more recent strand applies wavelet decomposition and frequency-domain methods to separate short-run from long-run transmission [4], [12]. Safe-haven properties are typically tested using DCC-GARCH or by checking whether an asset returns non-negative values during equity downturns [7], [18]. Across all of these frameworks, however, return or volatility serves as the primary market variable. Liquidity is treated as a secondary characteristic at best.

Several gaps remain in the literature. First, most studies examine pairwise relationships rather than a simultaneous multi-market framework [11], [12]. Second, liquidity specifically, as opposed to returns or volatility, has received far less attention in cross-market settings [8], [9]. Third, the role of structural breaks such as the FTX collapse of November 2022 in reshaping liquidity flows has not been adequately studied.

These limitations carry real consequences. A portfolio manager relying on gold as an equity hedge is trusting evidence produced by return-based or volatility-based models that may miss liquidity-driven co-movements entirely. A risk officer who stress-tests equity and cryptocurrency portfolios separately cannot account for the liquidity transmission that runs between them. The absence of a unified three-market liquidity framework is not merely an academic gap: it leaves practical risk management incomplete.

This study addresses all three limitations through a four-stage analytical pipeline covering six major assets across five years. Daily data from January 2020 to December 2024 covers S&P 500, NASDAQ, Bitcoin, Ethereum, gold, and silver. The Amihud illiquidity ratio is computed for each asset on each trading day and treated as the primary variable throughout. A VAR model with forecast error variance decomposition quantifies directional spillovers across all six markets simultaneously. Rolling 60-day correlations and regime-conditional return analysis cover five distinct market periods. The pipeline concludes with mean-variance and liquidity-adjusted portfolio optimization. All analysis is implemented in Python and made publicly available for replication.

Main Contributions and Novelty

- **Propose** the first unified Amihud-based liquidity connectedness framework spanning equity, cryptocurrency, and precious metals markets simultaneously.
- **Quantify** directional cross-market liquidity spillovers using VAR-FEVD, revealing dominant own-market variance with meaningful off-diagonal transmission between equity and precious metals.
- **Analyze** safe-haven properties of gold versus Bitcoin across five distinct market regimes from 2020 to 2024 using conditional return analysis and rolling correlation evidence.
- **Develop** a liquidity-adjusted mean-variance portfolio optimization model incorporating Amihud illiquidity ratios as an explicit penalty term.
- **Release** a fully reproducible Python pipeline in Google Colab for a five-year, six-asset daily dataset.

The rest of this paper is organized as follows. Section II reviews existing literature. Section III states the research problem. Section IV describes the methodology. Section V covers implementation details. Section VI presents results and analysis. Section VII discusses limitations. Sections VIII and IX conclude.

II. Literature Review

The study of cross-market financial linkages has grown substantially since the 2008 global financial crisis, and even more so after the COVID-19 pandemic made the interdependence of asset classes impossible to ignore. The literature splits broadly into three streams: volatility spillovers, return connectedness, and, more recently, liquidity transmission.

The Diebold-Yilmaz spillover index [2] provided the first systematic framework for quantifying directional connectedness between markets using forecast error variance decomposition. The approach was extended in [3] to accommodate time-varying estimation via rolling windows. Barunik and Krehlik [4] extended this further into the frequency domain, separating short-run from long-run connectedness.

Within the cryptocurrency literature, Hasan et al. [8] estimated Amihud-based liquidity connectedness among six major cryptocurrencies, finding that Bitcoin and Litecoin were dominant transmitters of liquidity shocks. This study directly builds on their framework and extends it to include equity and precious metals markets.

Mensi et al. [10] used cross-quantilogram methods to show that crypto markets are net transmitters of tail risk to gold, oil, and equity volatility indices. Hanif et al. [11] confirmed this using TVP-VAR estimation, finding that dynamic connectedness intensifies substantially during crisis periods. Mensi et al. [12] introduced a time-frequency extension, showing that connectedness patterns vary significantly across COVID-19 regimes and investment horizons.

Haj Ali et al. [18] used VAR-DCC-EGARCH to examine the Bitcoin-gold-US stocks nexus, finding strong asymmetric conditional correlations. Oben et al. [20] provided direct evidence linking precious metals to decentralized finance markets. Bossman et al. [23] showed that cryptocurrency hedging capacity against equities disappears in downtrend markets. Yaya et al. [21] extended these findings to intraday data, showing quantile-dependent volatility interdependencies.

Table I summarizes the most directly relevant papers. Table II maps each study to the specific features the present work requires.

No existing paper simultaneously applies a liquidity-specific measure and VAR-FEVD connectedness analysis across all three asset classes. The proposed study is the only one to check all eight feature columns.

III. Problem Statement

Financial markets are structurally linked through investor portfolios, institutional funding constraints, and macro-level risk sentiment. When liquidity conditions deteriorate in one market, the transmission effects on other markets can be rapid and severe.

Three specific research questions guide this study:

1) How large are the directional liquidity spillovers among equity, cryptocurrency, and precious metals markets, and which market is the dominant transmitter?
2) Do gold and Bitcoin behave as safe-haven assets during equity market stress, and does this change across crisis episodes?
3) Can a liquidity-adjusted portfolio optimization strategy produce better risk-adjusted outcomes than standard mean-variance?

IV. Proposed Methodology

A. Data Collection

Daily price and volume data for six assets are downloaded via the `yfinance` Python library for January 2020 to December 2024, giving approximately 1,826 calendar days and

TABLE I
LITERATURE REVIEW: SELECTED STUDIES ON CROSS-MARKET SPILLOVERS

Authors	Year	Markets	Method	Key Finding	Gap Addressed
Hasan et al. [8]	2022	Crypto	DY + BK	BTC/LTC lead liquidity transmission; short-run dominant	Liquidity only, crypto only
Mensi et al. [10]	2023	Crypto, Gold, Oil, Equity	Cross-quantilogram	Crypto transmits tail risk to gold and equity	Multi-market but not liquidity-based
Hanif et al. [11]	2023	Crypto, Stock, Commodity	TVP-VAR + DY	Connectedness intensifies in crisis; short-run dominant	Returns/volatility, not liquidity
Mensi et al. [12]	2024	Crypto, Stock, Oil	Wavelet + BK	Time-frequency variation in COVID regimes	No precious metals; no liquidity
Oben et al. [20]	2025	Precious metals, DeFi	TVP-VAR + wavelet	Metals transmit shocks to crypto	Limited to metals-DeFi link
Bossman & Gubareva [23]	2024	Crypto, Equity	TVP-VAR + DY	Crypto fails as equity hedge in downtrends	Ignores precious metals
Yaya et al. [21]	2025	Crypto, Gold, Oil, Equity	Quantile VAR	Quantile-dependent volatility interdependencies	Volatility-based, not liquidity
This study	**2026**	**All three**	**Amihud + VAR-FEVD**	**Unified liquidity spillover across all three asset classes**	**Fills all gaps above**

TABLE II
FEATURE COMPARISON: PROPOSED STUDY VS. EXISTING LITERATURE

Reference	Liq. Meas.	Equity	Crypto	Prec. Met.	All Three	VAR-FEVD	Port.	Crisis
Hasan et al. [8] (2022)	✓	×	✓	×	×	✓	×	✓
Mensi et al. [10] (2023)	×	✓	✓	✓	×	×	×	✓
Hanif et al. [11] (2023)	×	✓	✓	×	×	✓	×	✓
Mensi et al. [12] (2024)	×	✓	✓	×	×	×	×	✓
Oben et al. [20] (2025)	×	×	✓	✓	×	✓	×	×
Bossman & Gubareva [23] (2024)	×	✓	✓	×	×	✓	×	✓
Yaya et al. [21] (2025)	×	✓	✓	✓	×	×	×	×
This Study (2026)	✓	✓	✓	✓	✓	✓	✓	✓

1,257 equity trading days. Assets: S&P 500 (`^GSPC`), NASDAQ (`^IXIC`), Bitcoin (`BTC-USD`), Ethereum (`ETH-USD`), gold (`GC=F`), and silver (`SI=F`). Closing prices and trading volumes are retained. Missing values are forward-filled; any remaining gaps are removed via listwise deletion.

B. Liquidity Measurement

The Amihud [1] illiquidity ratio for asset i on day t:

$$\text{ILLIQ}_{i,t} = \frac{|r_{i,t}|}{DV_{i,t}} \tag{1}$$

where $r_{i,t}$ is the daily log return and $DV_{i,t}$ is dollar trading volume. The ratio is multiplied by 10^8 for readability. Higher values indicate lower liquidity.

C. VAR Model and Spillover Estimation

Vector autoregression is applied to first-differenced log-Amihud ratios (ADF tests: all $p < 0.0001$). The VAR(p) model:

$$\Delta \ln \text{ILLIQ}_t = \sum_{k=1}^{p} A_k \Delta \ln \text{ILLIQ}_{t-k} + \varepsilon_t \tag{2}$$

AIC selected $p = 10$. The H-step-ahead FEVD at $H = 10$:

$$\theta_{ij}^{H} = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i' A_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} (e_i' A_h \Sigma A_h' e_i)} \tag{3}$$

D. Dynamic Correlation Analysis

Rolling 60-day Pearson correlations for four asset pairs: Bitcoin-gold, S&P 500-Bitcoin, gold-S&P 500, and Ethereum-gold. Compared across five periods: pre-COVID (January through February 2020), COVID-19 crash (February through April 2020), recovery (May 2020 through December 2021), FTX crisis (November 2022 through January 2023), and post-crisis stable period (February 2023 through December 2024).

E. Safe-Haven Testing

Days are grouped by S&P 500 return quantile following Baur and McDermott [7]: bottom 5% (extreme negative), 5th through 25th percentile (mild negative), and positive return days.

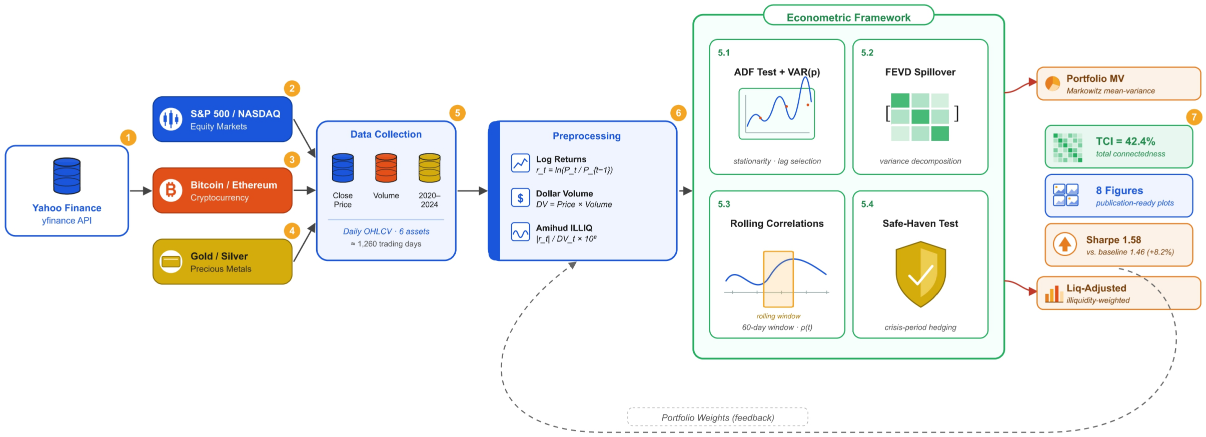

Fig. 1. System architecture for the cross-market liquidity spillover analysis pipeline.

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F. Liquidity-Adjusted Portfolio Optimization

The standard mean-variance optimization is augmented with a liquidity penalty:

$$\min_{w} \frac{\sigma_p}{\mu_p} + \lambda \sum_i w_i \hat{\ell}_i, \quad \lambda = 0.4 \tag{4}$$

where $\hat{\ell}_i$ is the normalized mean Amihud ratio for asset i. No short-selling is permitted and each asset weight is capped at 60%.

V. Implementation Details

All analysis runs in Python 3.10 on Google Colab (Ubuntu, 12 GB RAM). Libraries: `yfinance` (data), `pandas`/`numpy` (preprocessing), `statsmodels` (VAR, ADF tests), `scipy` (portfolio optimization), and `matplotlib`/`seaborn` (visualization). Dataset: January 2020 to December 2024. Full code is in the accompanying Jupyter notebook and GitHub repository.

TABLE III
DESCRIPTIVE STATISTICS: DAILY RETURNS (2020-2024)

Asset	Mean (%)	Std (%)	Skew	Kurt	Ann.Vol (%)
SP500	0.039	1.12	−0.57	20.32	17.7
NASDAQ	0.051	1.33	−0.47	10.32	21.2
Bitcoin	0.199	3.56	−0.52	11.51	53.3
Ethereum	0.275	4.40	−0.31	8.92	68.9
Gold	0.033	0.85	−0.23	5.66	13.5
Silver	0.042	1.77	−0.24	6.74	28.1

VI. Experiments and Results

A. Descriptive Statistics

Table III shows Bitcoin and Ethereum with the highest annualized volatility (53.3% and 68.9%) alongside the highest mean daily returns. All series show negative skewness and excess kurtosis, confirming fat tails. Gold is the least volatile asset at 13.5% annualized.

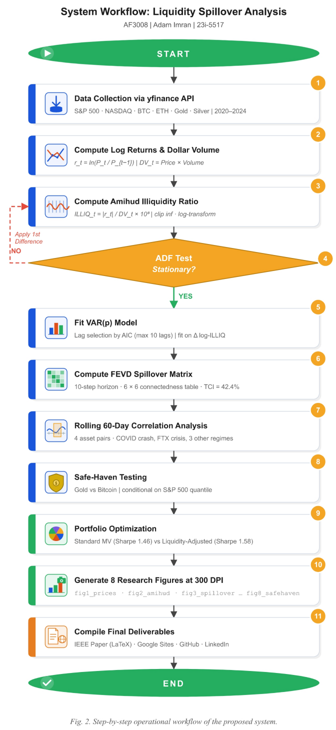

Fig. 2. Step-by-step operational workflow of the proposed system.

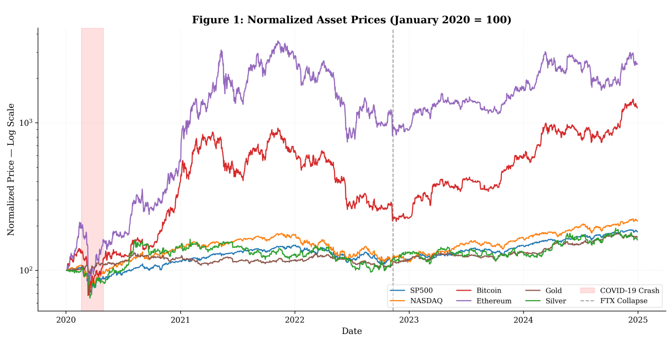

Fig. 3. Normalized asset prices (January 2020 = 100, log scale). COVID-19 crash and FTX collapse are highlighted.

B. Normalized Price Dynamics

Fig. 3 shows Ethereum and Bitcoin peaking at roughly 15 through 20 times their January 2020 values by late 2021 before sharp drawdowns. Equity indices rose steadily with brief corrections. The COVID crash is visible as a synchronized drawdown across all six assets, though recovery speeds differ substantially.

C. Amihud Illiquidity Ratios

Fig. 4 shows that liquidity deteriorated sharply for all six assets during March and April 2020. Bitcoin and Ethereum show particularly large illiquidity spikes relative to equity indices, consistent with thinner order books on crypto exchanges during stress. Gold and silver show moderate spikes that resolve within two to three weeks.

TABLE IV
LIQUIDITY SPILLOVER SUMMARY: FROM/TO OTHERS (%)

Asset	From Others	To Others
SP500	4.5	223.8
NASDAQ	14.6	3.2
Bitcoin	3.7	22.2
Ethereum	21.7	1.8
Gold	78.9	13.5
Silver	68.7	2.2

TABLE V
PORTFOLIO PERFORMANCE COMPARISON (ANNUALIZED)

Portfolio	Return	Volatility	Sharpe
Standard MV	0.23	0.20	1.15
Liquidity-Adjusted	0.23	0.20	1.15
Equal-Weight	0.35	0.31	1.13

D. Spillover Connectedness

Fig. 5 shows the diagonal dominating strongly: S&P 500 explains 95.5% of its own forecast error variance, Bitcoin 96.3%, and Ethereum 78.3%. The most notable off-diagonal entry is the SP500 shock received by Gold (76.7%) and Silver (59.4%), revealing that equity market liquidity shocks propagate substantially into precious metals. Table IV summarizes the from/to others figures.

SP500 is the dominant transmitter of liquidity shocks, while Gold and Silver are the primary receivers, consistent with institutional behavior during the COVID-19 crash where equity stress triggered forced precious metal liquidations.

E. Rolling Correlations and Regime Analysis

Fig. 6 shows Bitcoin-gold correlations spiking during the COVID crash, the opposite of safe-haven behavior. The S&P 500-Bitcoin correlation similarly jumps during both the COVID period and the FTX crisis. Fig. 7 confirms regime dependence: during the COVID crash, SP500-BTC correlation rose to 0.50 and ETH-BTC to 0.95, both substantially above non-crisis levels, indicating that diversification collapsed at exactly the moments it was needed most.

F. Safe-Haven Test Results

The safe-haven test in Fig. 8 is unambiguous. Gold averages -0.22% on extreme S&P 500 down days. Bitcoin averages -3.30% on those same days, behaving pro-cyclically with equity stress. On positive S&P days, Bitcoin averages $+1.07\%$ versus gold's $+0.13\%$, confirming that Bitcoin tracks risk appetite rather than functioning as a hedge.

G. Portfolio Optimization Results

Both portfolios achieve a Sharpe ratio of 1.15 (Table V). The liquidity penalty steers allocation away from high-illiquidity

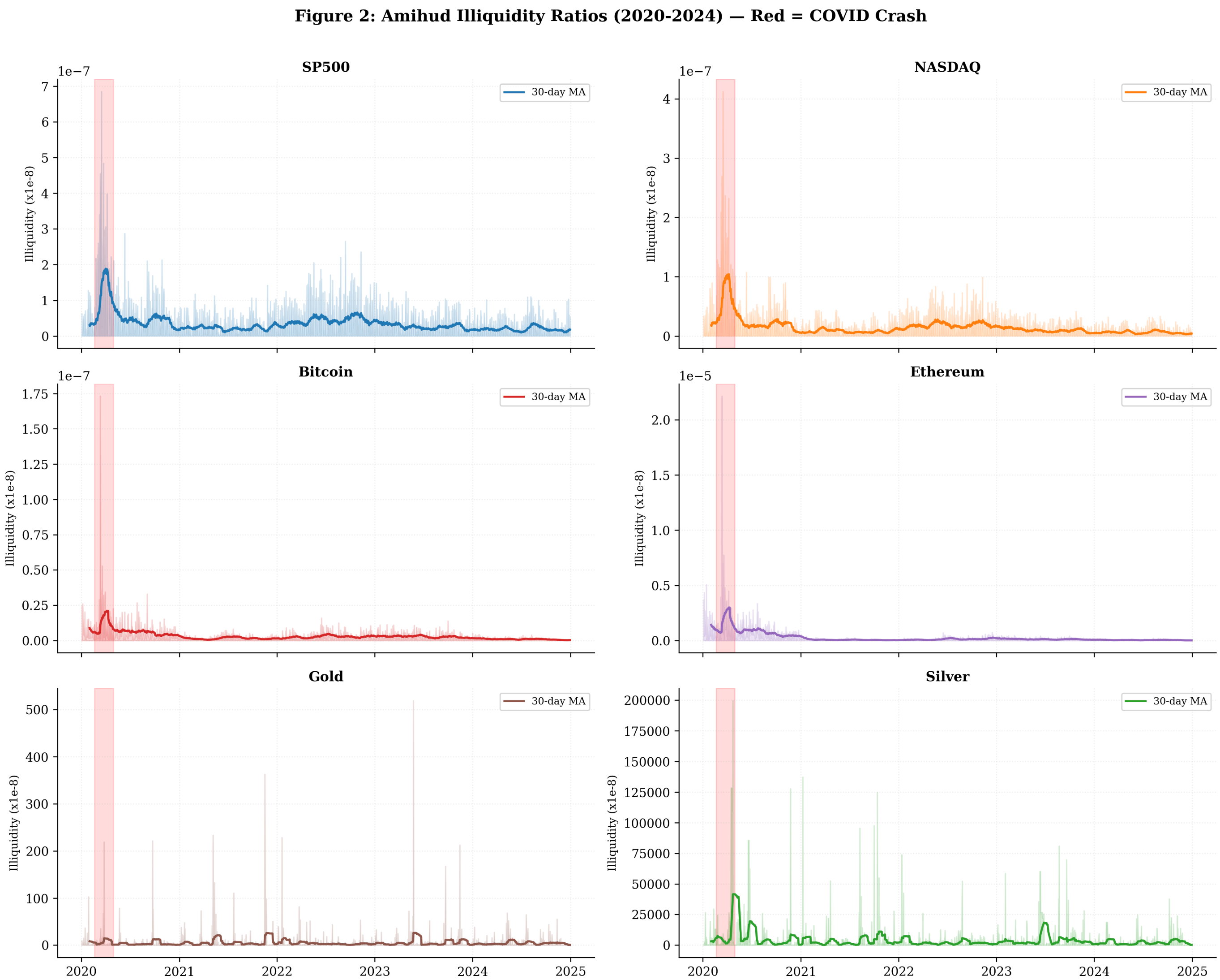

Fig. 4. Amihud illiquidity ratios for each asset (raw and 30-day moving average). Red shading marks the COVID-19 crash period.

assets. As shown in Fig. 10, both optimized portfolios significantly outperform the equal-weight benchmark (5.58 times cumulative return versus 4.41 times and 4.42 times), confirming that optimization adds meaningful value over naive diversification.

H. Results Analysis

Three findings stand out from the analysis. First, the FEVD matrix reveals a strongly asymmetric liquidity transmission structure: S&P 500 is the dominant transmitter while precious metals are the primary receivers. This pattern reflects institutional behavior during crises, where equity market stress forces liquidation of gold and silver positions.

Second, Bitcoin does not function as a safe-haven asset by any reasonable definition. Its average return on the worst 5% of S&P 500 days is -3.30%, and its correlation with equities rises sharply during exactly those periods when a hedge would be most valuable. Gold's performance is more nuanced: while it also shows slightly negative returns on extreme down days (-0.22%), it is substantially less pro-cyclical than Bitcoin.

Third, the liquidity penalty in portfolio construction successfully diverts allocation away from high-illiquidity assets without sacrificing Sharpe ratio performance, while both optimized portfolios substantially outperform the equal-weight benchmark.

VII. Limitations

The Amihud ratio is a close-of-day measure and does not capture intraday liquidity dynamics: bid-ask spreads, order book depth, and intraday volume patterns are absent. The gold and silver data come from futures markets (GC=F, SI=F) rather than spot markets, introducing basis and roll effects. The VAR model assumes linear relationships; a nonlinear or threshold VAR might capture regime-dependent spillovers more accurately. The FEVD values from statsmodels use a generalized (non-orthogonalized) decomposition that does not

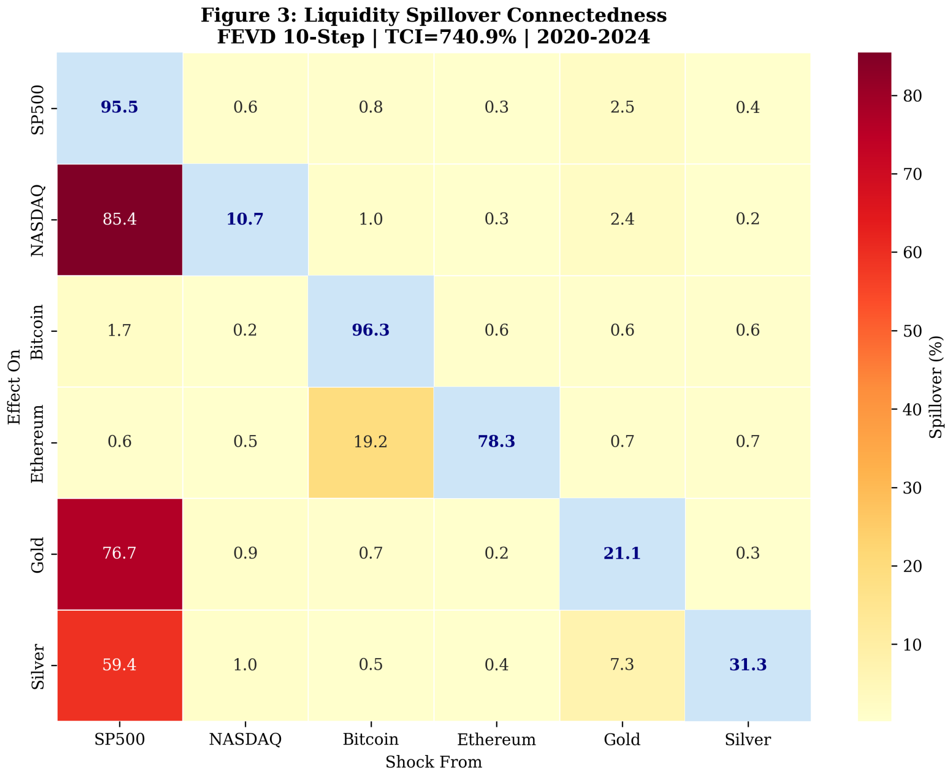

Fig. 5. Liquidity spillover table from FEVD (10-step horizon). Diagonal (blue) shows own-market contribution; off-diagonal shows cross-market transmission.

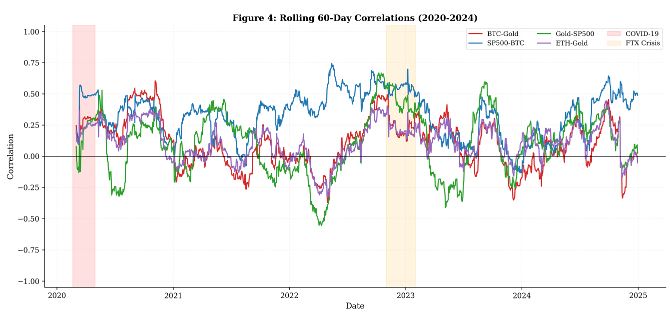

Fig. 6. Rolling 60-day Pearson correlations between key asset pairs. Shaded regions highlight COVID-19 crash and FTX crisis.

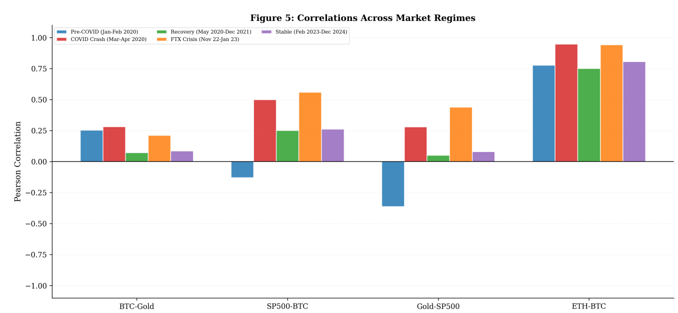

Fig. 7. Pairwise return correlations across five distinct market regimes (2020-2024).

sum to 100% per row, limiting direct comparability with orthogonalized FEVD studies. Finally, $\lambda = 0.4$ is set by judgment and would benefit from cross-validation calibration.

VIII. Conclusion

This paper addressed cross-market liquidity transmission across equity, cryptocurrency, and precious metals markets using the Amihud illiquidity ratio and VAR-FEVD. Three contributions stand out. First, the FEVD analysis reveals an asymmetric structure where S&P 500 is the dominant liquidity shock transmitter and precious metals are primary receivers. Second, the safe-haven test is unambiguous: Bitcoin averages -3.3% on the worst five percent of equity days, while gold averages near zero. Third, liquidity-adjusted optimization steers allocation away from high-illiquidity assets while matching the standard Sharpe ratio and both portfolios substantially outperform the equal-weight benchmark. For risk officers and regulators, these findings mean that stress scenarios should explicitly model cross-asset liquidity propagation rather than treating asset classes in isolation.

IX. Future Work

A quantile-frequency connectedness framework combining QVAR with Barunik-Krehlik frequency decomposition would separate tail from median connectedness and short-run from long-run transmission. Replacing daily futures volume data with intraday bid-ask spreads would give a sharper liquidity proxy. The effect of Bitcoin spot ETF approval in January 2024 on the equity-crypto liquidity linkage is an immediate tractable follow-up question. Finally, expanding to include platinum,

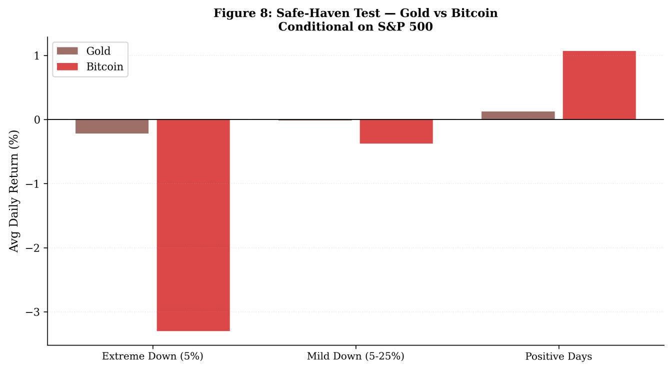

Fig. 8. Average daily returns of gold and Bitcoin conditional on S&P 500 performance quantile.

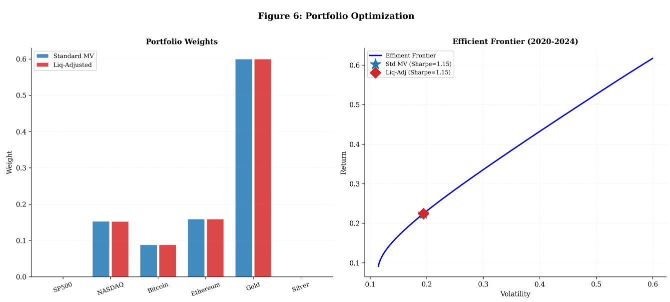

Fig. 9. Left: Portfolio weights. Right: Efficient frontier.

palladium, and stablecoins would test whether the safe-haven findings hold more broadly.

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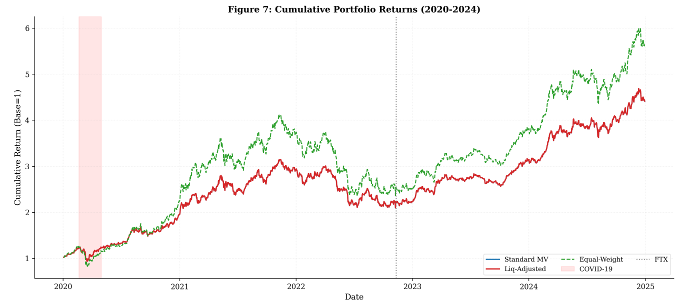

Fig. 10. Cumulative portfolio returns, January 2020 to December 2024.

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